

Exam Type: Practical
Duration: 3 Hours (180 Minutes) | Maximum Marks: 50
Data Science & AI/ML
Set A

All five modules are compulsory.

Predict campaign responses and identify audience segments.

Technical: 25 marks | Video: 15 marks | GitHub: 10 marks

Exam Instructions

Work individually on your assigned set. Use the supplied data generator unchanged. Cite external code or references and follow the invigilator's resource-use rules. Complete the analysis, recording and submission within 180 minutes.

Use Python for all five modules. One executed notebook with five clearly labeled sections is sufficient. Save editable code and outputs; screenshots alone are insufficient. No deployment, GPU or exhaustive model tuning is required.

Suggested Time Allocation

Phase	Time	Focus
Data setup	10 min	Generate and inspect the dataset
Maths & Stats	25 min	Statistics, inference and matrix calculation
Data preparation	30 min	Cleaning, split and fitted transformations
Supervised	30 min	Baseline, classifier and evaluation
Unsupervised	20 min	Clustering
ANN	25 min	Dense network and learning curves
Video recording	15 min	5–10 minute face and screen explanation
GitHub & final checks	25 min	README, commits, links and reproducibility
Total	180 min	All work included

Lab Prerequisites

Before the timer: Python with NumPy, pandas, SciPy, scikit-learn, matplotlib and TensorFlow/Keras (or PyTorch); a notebook/editor; GitHub access; screen and webcam recorder. The invigilator must verify that a small ANN runs on the lab CPU. Installation is not assessed.

Dataset & Calculation Rules

Create `src/generate_data.py` from the block below and run it once from the repository root. It creates `data/raw/set_d.csv` with 305 rows, including five exact duplicates. The clean dataset has 300 distinct records. This supplied setup code earns no solution marks.

```
from pathlib import Path
import numpy as np
import pandas as pd
rng = np.random.default_rng(404)
n = 300
cols = ['visits', 'recency', 'engagement', 'spend']
x = rng.normal(size=(n, 4))
group = rng.choice(["G1", "G2"], size=n)
score = x @ np.array([0.6, -0.9, 1.1, 0.4])
score += 0.4 * (group == "G2") + rng.normal(0, 1, n)
y = (score > 0).astype(int)
df = pd.DataFrame(np.round(50 + 10*x, 2), columns=cols)
df.insert(0, "record_id", np.arange(1, n+1))
df["group"] = group
df["response"] = y
for col in cols[:2]:
    df.loc[rng.choice(n, 15, replace=False), col] = np.nan
df = pd.concat([df, df.iloc[:5]], ignore_index=True)
Path("data/raw").mkdir(parents=True, exist_ok=True)
df.to_csv("data/raw/set_d.csv", index=False)
```

Data Dictionary and Required Feature

Numeric predictors: visits, recency, engagement, spend. These are synthetic, dimensionless index measurements, not physical units. group is a categorical operational cohort (G1/G2). record_id is an identifier; exclude it from every model. response is the binary target: 1 means response, 0 means its absence.

Engineer one predictor named `engineered_feature`: $\text{engagement} / (\text{recency} + 1)$. Calculate it after numeric imputation, using original-scale values before scaling. Never include the target in the feature calculation.

Keep raw CSV unchanged. Remove exact duplicates before splitting. Use `random_state=42` for an 80/20 stratified train/test split. Reserve 20% of the training partition for ANN validation, stratified with `random_state=42`. Save record IDs for each partition. The expected sizes are 192 fit, 48 validation and 60 test records.

For a fair comparison, fit imputers, encoders, scalers and both predictive models on the same 192 fit records. Use the 48 validation records only for ANN stopping. Evaluate the final models once on the same untouched 60 test records. Use seed 42 for model randomness; record library versions.

Section A — Technical Work (25 Marks)

Task 1 — Maths & Adv Stats (5 Marks)

Use only the 192 fit records after deduplication. For statistical calculations, use observed values rather than imputed values and report the sample size used.

M1 — Descriptive Statistics (1 Mark)

- For engagement, report observed n, mean, median and sample standard deviation (ddof=1). Save a histogram with title and axis labels. Correct summary: 0.5; readable histogram: 0.5.

M2 — Statistical Inference (2 Marks)

- Compare mean engagement between G1 and G2 with a two-sided Welch t-test at $\alpha=0.05$. State H_0 and H_1 , report group counts, test statistic and p-value (1). Report a 95% t-confidence interval for the overall observed mean of engagement, state independence/approximate-normality assumptions, and interpret both results without claiming causation (1).

M3 — Linear Algebra (2 Marks)

- Using fit rows complete for engagement and visits, manually center the two-column matrix. Compute its sample covariance matrix as $X_centered.T @ X_centered / (n-1)$ (1). Obtain eigenvalues with a symmetric-matrix routine and report the largest eigenvalue divided by their sum, explaining the corresponding principal direction and variance share (1).

Task 2 — Data Preprocessing & Feature Engineering (5 Marks)

F1 — Audit and Partition (2 Marks)

- Report shape, exact duplicates, missing counts and target counts; verify 300 unique records after cleaning (1). Create the specified fit/validation/test partitions, save their IDs, and prove they are disjoint (1).

F2 — Transform and Engineer (2 Marks)

- Fit median imputation on numeric fit columns and one-hot encode group, handling unseen categories; transform validation/test without refitting (1). Create the specified engineered feature, then fit standard scaling on fit numeric features only; reuse it unchanged elsewhere (1). Keep group one-hot columns unscaled.

F3 — Leakage Evidence (1 Mark)

- Print transformed shapes and feature names; verify finite numeric outputs. Explain why the target, record_id and test statistics are excluded from training. Save fitted preprocessing or provide executable code that recreates it.

Technical Work — Predictive Models and Segments

Task 3 — Supervised Learning (5 Marks)

S1 — Baseline and Classifier (2 Marks)

- Fit a majority-class DummyClassifier on the fit partition (0.5). Train LogisticRegression with max_iter=1000 on the transformed fit features (1). Document the target, predictors, model settings and fixed split IDs (0.5).

S2 — Holdout Evaluation (2 Marks)

- On the untouched test set, report baseline accuracy and classifier accuracy, precision, recall and F1 for positive class 1 (1). Save a labeled confusion matrix and per-record test predictions including record_id, true label, predicted label and class-1 probability (1). Use threshold 0.5 and document handling of undefined precision/recall.

S3 — Interpretation (1 Mark)

- Explain a false-positive versus false-negative consequence for this scenario. Use the test metrics and baseline to justify whether the classifier offers useful improvement. High accuracy is not a prerequisite for marks.

Task 4 — Unsupervised Learning (5 Marks)

U1 — K-Means and Selection (3 Marks)

- Use the transformed numeric fit predictors plus engineered_feature; exclude group one-hot columns, target and ID. Fit KMeans for k=2,3,4 with n_init=10 and random_state=42. Report inertia and silhouette score (1). Choose the highest-silhouette k; break ties with the smaller k and explain the choice (1).

U2 — Segment Interpretation (2 Mark)

- Give each cluster a descriptive name based on its profile and propose one practical action. Explain why cluster IDs are arbitrary and do not represent target classes. Do not use test data or target labels to choose clusters.

Save model metrics and clustering diagnostics as CSV or JSON in outputs/. Save plots in outputs/figures/. Keep all calculations in the submitted notebook or scripts.

Technical Work — Deep Learning up to ANN

Task 5 — Deep Learning up to ANN (5 Marks)

D1 — ANN Architecture and Compilation (2 Marks)

- Implement a dense feed-forward ANN using TensorFlow/Keras or equivalent PyTorch: input width equals transformed feature count; hidden layers of 16 and 8 ReLU units; one sigmoid output. Display model summary and count trainable parameters (1). Use binary cross-entropy loss and Adam optimizer at learning rate 0.001; explain the output/loss choice (1).

D2 — Training and Validation (2 Marks)

- Train on the 192 fit records with batch size 16 for at most 50 epochs. Use the reserved 48 validation records and early stopping on validation loss with patience 5, restoring best weights. Record the actual epochs and random seed (1). Plot training and validation loss against epoch and discuss signs of underfitting or overfitting (1).

D3 — Comparison and Evidence (1 Mark)

- Evaluate the restored ANN once on the same 60 test records using threshold 0.5. Report accuracy, precision, recall and F1, and compare F1 with logistic regression. Save ANN predictions and trained weights/model in models/. State a reasoned model preference; the ANN need not outperform logistic regression.

Evidence and Reproducibility

Save the executed notebook with visible outputs or runnable scripts plus saved result files. Document how to load the saved ANN and preprocessing, or how to reproduce training. Use relative paths. A CPU is sufficient for this dataset and architecture.

Minimum result files: statistics summary and inference results; split IDs and preprocessing audit; baseline/logistic/ANN metric comparison; classifier confusion matrices and predictions; k-selection scores and cluster profiles; histogram, and ANN loss curves.

Interpretation Rules

Treat all data as synthetic practice observations. Report statistical uncertainty, sample sizes and the limits of a single small holdout. Do not claim causal effects or deployment readiness. Never tune on the test set or choose a random seed to improve a reported score.

Technical marking principle

Each module carries 5 marks. Correct methods, reproducible evidence and clear interpretation earn credit; no minimum prediction score is imposed.

Section B — Project Explanation Video (15 Marks)

Record one video lasting 5–10 minutes, with a maximum of 10 minutes. Keep your face visible in a webcam overlay alongside the screen recording throughout. Use clear voice narration and readable screen content. Explain your own work rather than reading the question paper.

Suggested Recording Sequence

- Introduce your name, student ID, Set A and the campaign response problem.
- Explain dataset generation, duplicate handling, missing data and the fit/validation/test split.
- Show the hypothesis test, confidence interval and covariance/eigenvalue calculation.
- Explain fitted preprocessing and engineered_feature; point out how leakage is prevented.
- Show logistic regression metrics, the baseline and the confusion matrix.
- Explain k selection, and cluster profiles.
- Show ANN architecture, loss curves and comparison with logistic regression.
- Conclude with two numerical findings, a recommendation, a limitation and a repository walkthrough.

Marking Criteria

Criterion	Marks and evidence
V1 — Problem, data and preparation	3: problem (1), data structure (1), cleaning and split decisions (1).
V2 — Technical explanation	6: each of the five modules (1 each), plus clear end-to-end workflow and leakage explanation (1). Show output and explain logic.
V3 — Interpretation	3: two correct numeric findings (1 each); recommendation with limitation (1).
V4 — Presentation	3: face and screen visible throughout (1); audio and screen clarity (1); duration 5–10 minutes (1).

Video Hosting & Submission

Host the video where the examiner can view without requesting access, such as an unlisted YouTube video or an accessible Drive video link. Put its URL and duration in README.md. Test playback while signed out. The video itself need not be committed to GitHub. An inaccessible video scores 0 for Section B.

Section C — Public GitHub Submission (10 Marks)

Create a public repository named ds-aiml-set-d-YOUR-STUDENT-ID, replacing the final placeholder with your identifier. Put your name, ID and assigned set in README.md. Submit the repository URL and final commit hash to the examiner within 180 minutes.

Required Repository Structure

File / Path	Purpose
README.md	Identity, objective, setup, findings and video link
requirements.txt	Python package versions
.gitignore	Exclude environments, caches and secrets
src/generate_data.py	Supplied generator, unchanged
data/raw/set_d.csv	Original generated data with duplicates and missing values
notebooks/exam.ipynb	Executed work with five module sections; src/ scripts also acceptable
outputs/splits.csv	record_id and fit/validation/test membership
outputs/	Statistics, audits, metrics, predictions and cluster profiles
outputs/figures/	Histogram, confusion matrices, and loss curves
models/	Saved ANN and fitted preprocessing, or documented preprocessing recreation

README.md Must Include

- Project title, student name/ID, set, objective and data dictionary.
- Generator command, cleaning rules, feature formula, split seed and partition sizes.
- Tools/package versions, folder map and exact setup/run sequence from repository root.
- Statistical hypotheses, assumptions, confidence interval interpretation and key outputs.
- Model settings, held-out comparison, cluster findings, two numerical findings and one recommendation with a limitation.
- Reconciliation: test metrics must match saved predictions and use identical test IDs for both models.
- Accessible video URL and duration, references, and declaration: All work is my own except where cited.

Submission Rubric and Final Checks

Criterion	Marks and evidence
G1 – Public repo and deliverables	2: signed-out public link (1); source files and outputs present/openable (1).
G2 – Project organisation	2: clear folders and filenames (1); requirements.txt and .gitignore (1).
G3 – README.md	4: identity/objective/data (1); setup and run steps (1); findings and prediction reconciliation (1); video link, duration and authorship/references (1).
G4 – Reproducible handoff	2: documented execution works on another machine (1); meaningful commits and submitted final hash (1).

Marks Summary

Component	Marks	Breakdown
Maths & Adv Stats	5	M1 1 + M2 2 + M3 2
Preprocessing & Feature Engineering	5	F1 2 + F2 2 + F3 1
Supervised Learning	5	S1 2 + S2 2 + S3 1
Unsupervised Learning	5	U1 2 + U2 2 + U3 1
Deep Learning up to ANN	5	D1 2 + D2 2 + D3 1
Technical subtotal	25	Five compulsory modules
Video explanation	15	V1 3 + V2 6 + V3 3 + V4 3
GitHub submission	10	G1 2 + G2 2 + G3 4 + G4 2
GRAND TOTAL	50	

Final Checks Before Submission

- Public repository and video open while signed out.
- Raw file preserved; 300 unique records; split IDs are disjoint.
- Transforms fitted only on fit data; both models use identical holdout IDs.
- Notebook runs from a clean kernel; outputs and saved ANN are included.
- README has setup instructions, findings, video duration and authorship.
- Final repository URL and commit hash submitted before the deadline.

Award partial credit within each stated criterion for demonstrably correct work. Do not add unspecified deductions. Never commit passwords, tokens or unrelated personal data.

BRING ON YOUR CODING ATTITUDE